ANALYSIS OF TRAVEL TIMES

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Significance Of Travel Time Reliability

- Travel time reliability is the consistency or dependability in travel times, as measured from day-to-day and/or across different times of the day.
- Travel time reliability & variability can be used interchangeably.The larger the travel time variability is, the worse the travel time reliability is.
- Reliability is highly important because travelers must either budget extra time for their trips to avoid arriving late or suffer the consequences of being late.
- Factors causing travel times to be unreliable are : Incidents , inclement weather, work zones , special events etc.
- Reliability can only be characterized by a long history of travel times, and the use of automated equipment is the only feasible method of data collection.

Reliability Metrics

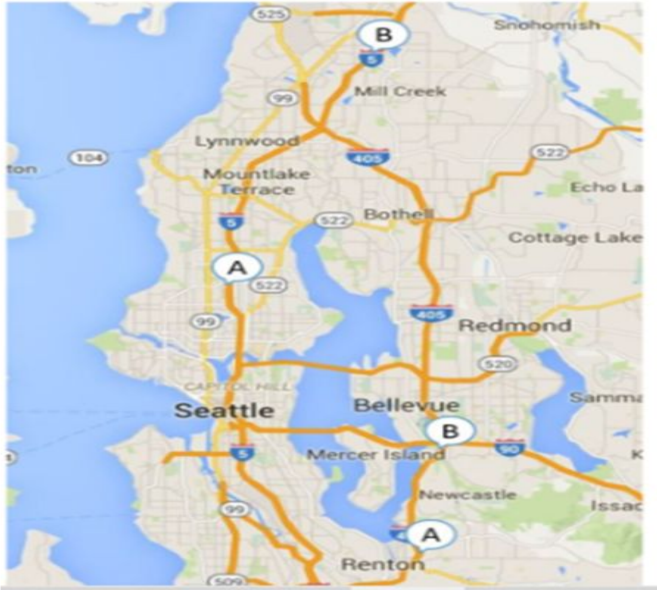
- Reliability metrics provide an understanding of how dependable or variable travel conditions are, but they do not identify the cause of the variability. In this sense, reliability measures are top-level outcome measures.

  Some travel time reliability metrics are :

- The **Travel Time Index** (TTI) is the ratio of the travel time during the peak period to the time required to make the same trip at free-flow speeds.
- The **Planning time index** is computed as the 95th percentile TTI.
- The **Buffer time index** is difference between 95th percentile TTI and average TTI , normalized by average TTI.

Impact Of Incidents

- The study area was located in the Seattle metropolitan area, Travel time reliability of Interstate 5 was examined.
- This study aims to examine the impact of traffic incidents occurring on general purpose lanes (GPLs) and high-occupancy vehicle (HOV) lanes on freeways and to evaluate the differences of travel time reliability on GPLs and HOV lanes under the same incident conditions.



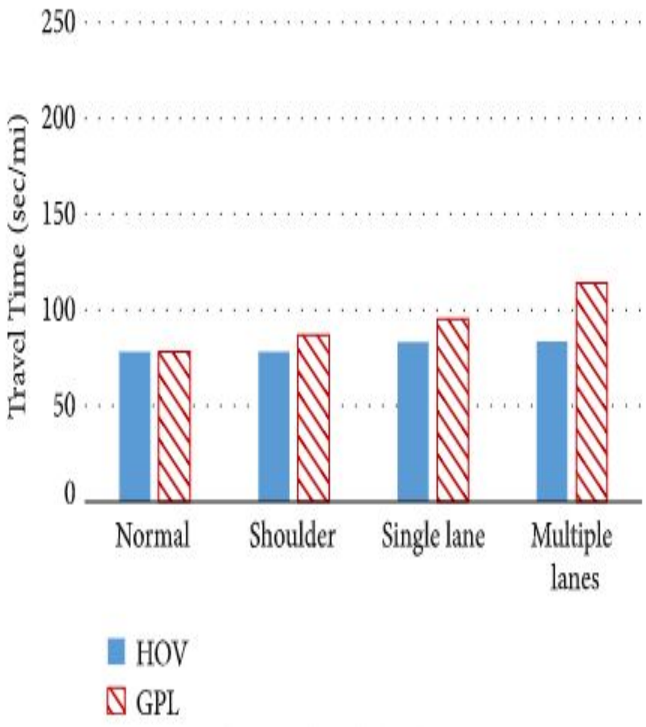


Figure: Comparison of 90th percentile travel time between HOV lanes and GPLs on I-5 South

Inferences:

- Travel time variability of I-5 South is analysed under four conditions.
- The difference of travel time reliability between multiple lane incidents and normal situation is 28.17%(on HOV lanes)

Weather-Related Statistics

- Weather acts through visibility impairments, precipitation, high winds, and temperature extremes .
- According to Road Weather Management Program by U.S. Department Of Transportation Federal Highway Administration, On average, there are over 5,891,000 vehicle crashes each year(Ten year average from 2007-2016). Approximately 21% of these crashes are weather-related.

Weather conditions	Speed reduction	Volume Reduction	Capacity Reduction
light rain/snow	2%-13%	5%-10%	4%-11%
heavy rain	6%-17%	14%	10%-30%
heavy snow	5%-64%	30%-44%	12%-27%

Table:Freeway traffic flow reductions due to weather

Impacts Of Weather & Crashes

- The effects of crashes and other kinds of traffic disruptions are of significant interest because they are common causes of travel delay .
- To examine these impacts, a detailed analysis is performed using data from seattle to measure roadway performance data(volumes & travel times taken every 5 mins.)
- Because each 5-minute time period is different on different days.For example,the 7:00 a.m. travel time today has no influence on the 7:00 a.m. travel time tomorrow.
- But the drawback is it is required to perform many statistical tests.To reduce the analytic load, the project team grouped the 5-minute average travel times by 30-minute increments, with the statistical tests performed for each 30-minute interval.
- For example,assume that no incident happens in a study corridor on a particular day until 7:15 am .All three 5-minute average travel times are included in the computation of the travel time distribution for the not incident–influenced 30-minute period covering 7:00 to 7:29 a.m., and the three 5-minute periods from 7:15 to 7:25 a.m. are included in the incident influenced travel time distribution for that same 7:00 to 7:29 a.m. period.

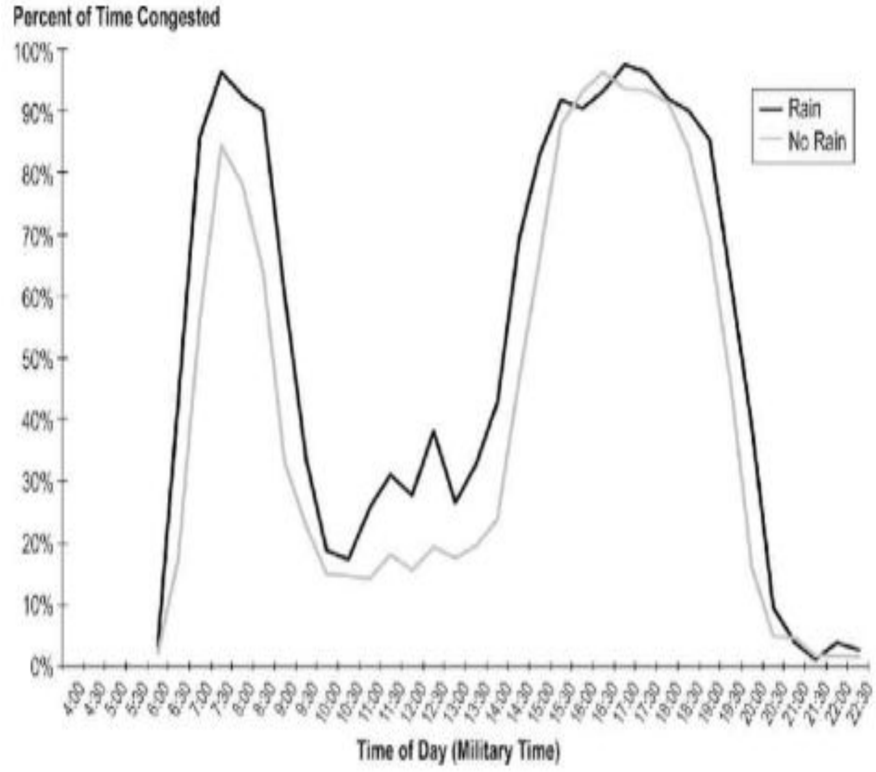


Figure : Probability of being in congestion: rain versus no rain on SR 520 westbound from Bellevue to Seattle.

Inferences:

- The results uniformly showed that the occurrence of rain led to a statistically significant increase in the amount of congestion, but only during periods of moderately high traffic volume.
- That is, rain does not cause congestion uniformly throughout the day. The probability of congestion forming as a result of rain is a function of the underlying level of vehicular demand.

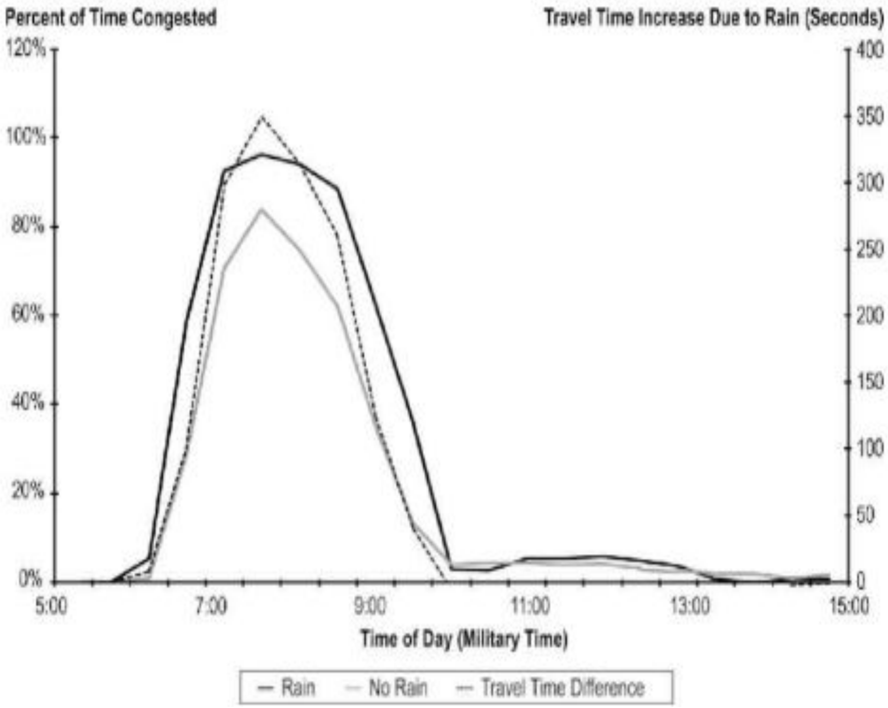


Figure :Correspondence of increase in mean travel times with increase in probability of congestion due to rain on I-90 westbound from Issaquah.

Inferences:

- Figure illustrates how mean travel times increase along with the increased probability of being in congestion..
- It shows at no time does the mean travel time decrease with statistical significance when rain is present.
- Interestingly, this figure also shows that the declining volumes at the end of the commute period quickly moderate the travel time effects of the congestion..

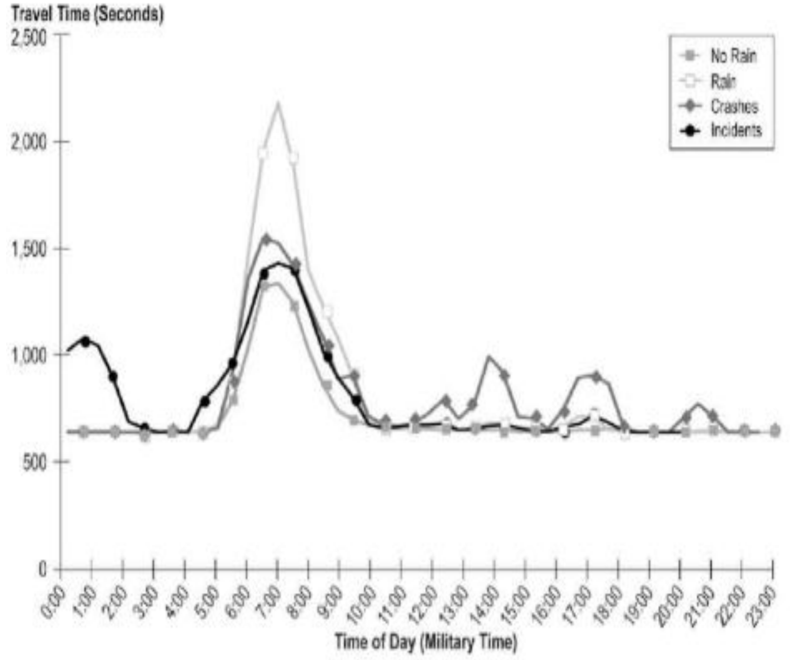


Figure:Mean Travel Time under rain,crash, or non crash traffic incident conditions on I-5 northbound ,south corridor

Inferences:

- The impacts of any disruption are a function of the underlying traffic volume condition during which that disruption occurs.
- The next most important factor is the size of the disruption imposed on the traffic stream.
- Therefore, crashes frequently have more significant effects during times of lower volume.That is, once the roadway congests, a large disruption occurring before congestion forms can create an even larger change in expected travel times during the course of the peak period because of the growth of the queue associated with the initial congestion point.

Congestion Analysis

- Reliability is an aspect of total congestion that is greatly influenced by the complex interactions of traffic demand, physical capacity, and roadway events.
- Congestion occurs when there is too much volume and too little roadway capacity. This can occur because Traffic demand is too great for the designed roadway capacity; or Some disruption reduces functional roadway capacity (supply) to levels below demand.

Source	Congestion (%)
Recurring (bottleneck)	48.4
Incidents	32.8
Weather	11.1
Incidents and weather	7.7

Table: Congestion by source during peak period in atlanta(2008)

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QUESTIONS?